

Erich MacLean

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TECHNICAL SKILLS

Languages: Python, Java, C, VHDL, JavaScript, Assembly, HTML/CSS, Bash, PowerShell

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-Learn, XGBoost, YOLO, OpenCV, ONNX, Pandas, NumPy

Tools & Platforms: Docker, AWS, Azure, Git, Tailscale, WireGuard, Supabase, RTOS (FreeRTOS), Arduino, FPGA

Areas: Machine Learning, Embedded Systems, Computer Vision, Digital Logic, Systems Administration

EXPERIENCE

Graduate Teaching Assistant | University of Guelph | Guelph, ON Sep 2023 - Dec 2025

- Delivered labs, tutorials, and office hours for Digital Systems Design and Computer Organization
- Produced supplementary learning materials covering VHDL, digital logic, and CPU architecture
- Graded assignments, reports and exams for 100+ students; nominated for TA of the Year (CEPS) in 2024 and 2025

IT Support Coordinator | D2L | Kitchener, ON May 2024 - Aug 2024

- Built a custom web app integrating Azure Cognitive Services image recognition to scan barcodes and execute bulk Jira API operations, reducing asset processing overhead across enterprise deployment workflows
- Deployed the application on Azure Web Services with domain-level, role-based access controls and transitioned user domain management scripts to serverless Azure Functions, eliminating persistent VM infrastructure costs
- Centralized migrated scripts in a D2L-managed GitHub repository to improve version control, and managed macOS/Windows device lifecycles (data migration, OS reinstallation, and factory resets)

Driving Simulator Lab Technician & Developer | University of Guelph | Guelph, ON May 2022 - Aug 2023

- Engineered a Python pipeline interfacing with the SCANer Studio API, eliminating a post-simulation export bottleneck (saving up to 2 hours per run) via live data capture during active simulation execution
- Designed a Tkinter GUI enabling non-technical researchers to operate the pipeline; collaborated with CS, Engineering, and Psychology research teams to design multi-scenario simulation environments and road hazard layouts
- Maintained a 9-PC networked simulation stack, troubleshooted hardware/software faults, and conducted vendor outreach and benchmarking with University of Waterloo and Toronto to inform hardware procurement strategies

PROJECTS

Collision Prediction Framework | Autonomous Vehicle Safety Systems May 2023 - Apr 2026

MASc Thesis Research, University of Guelph

- Developed multimodal ML classifiers (PyTorch, TensorFlow, YOLO, OpenCV) for real-time vehicle collision prediction using simulation driving and eye-tracking data, achieving up to 97% prediction accuracy
- Produced a generalizable evaluation framework to assess ML model reliability under real-world physical constraints, designed for ADAS and autonomous driving validation pipelines

Personal Home Server and Secure Mesh Network 2023 - Present

Personal Project

- Provisioned a headless Ubuntu home server on dedicated hardware to self-host secure file storage, media streaming, and containerized services
- Architected a private mesh network across daily devices using Tailscale, implementing per-device firewall policies to enforce zero-trust network access
- Engineered a CGNAT bypass using an Oracle Cloud VM as a relay node, enabling secure external access to home services without a public IP

EDUCATION

Master of Applied Science, Engineering (Artificial Intelligence) Sep 2023 - Apr 2026

University of Guelph, Guelph, ON

Bachelor of Engineering, Computer Engineering | (Dean's Honor List, Fall 2021) Sep 2019 - Apr 2023

University of Guelph, Guelph, ON